

ST NAZAIRE MONTOIR AP, France

WMO: 072170

Lat: 47.3139N

Lon: 2.1544W

Elev: 13

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	29.2	20.6	15.4	29.6	24.0	18.1	31.3	30.7	52.5	27.7	51.7	5.3	30	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	86.0	67.6	81.8	66.4	78.2	65.0	70.0	80.9	68.2	77.9	66.7	74.9	10.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	97.4	73.4	65.0	92.5	71.5	63.7	88.5	69.9	33.9	81.2	32.4	77.8	31.2	74.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.3	22.0	19.4														
			DB	22.5	92.4	3.0	4.0	20.4	95.3	18.6	97.7	17.0	99.9	14.8	102.9	
			WB	21.5	73.4	3.0	1.9	19.3	74.8	17.6	75.9	15.9	77.0	13.7	78.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	43.7	44.1	48.0	52.0	57.5	63.5	66.4	65.7	62.0	56.3	49.3	44.7	(6)
(7)		DBStd	9.91	6.33	5.86	5.15	4.98	5.05	5.46	4.86	4.74	4.91	5.42	5.90	6.77	(7)
(8)		HDD50	797	207	174	98	32	3	0	0	0	0	12	83	188	(8)
(9)		HDD65	4060	662	586	526	389	238	90	38	43	114	273	471	630	(9)
(10)		CDD50	2442	11	8	36	94	237	407	510	488	361	206	61	23	(10)
(11)		CDD65	229	0	0	0	0	7	46	83	66	25	2	0	0	(11)
(12)		CDH74	1909	0	0	0	6	76	424	684	508	204	7	0	0	(12)
(13)		CDH80	555	0	0	0	0	10	115	223	164	43	0	0	0	(13)
(14)	Wind	WSAvg	9.2	9.8	10.1	10.1	9.5	9.0	8.8	9.0	8.3	8.2	8.7	9.2	10.1	(14)
(15)	Precipitation	PrecAvg	30.9	3.5	2.7	2.2	2.3	2.2	1.5	1.7	1.6	2.4	3.5	3.7	3.6	(15)
(16)		PrecMax	39.7	7.0	6.0	6.3	6.3	4.6	3.1	3.7	4.3	6.8	7.3	7.4	6.3	(16)
(17)		PrecMin	20.2	0.4	0.2	0.2	0.6	0.5	0.2	0.2	0.1	0.1	0.7	1.1	0.2	(17)
(18)		PrecStd	5.0	1.9	1.5	1.5	1.4	1.0	0.9	0.8	1.1	1.6	1.7	1.8	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	59.2	68.1	74.3	81.1	89.6	91.5	91.2	85.5	74.1	64.1	58.4	(19)
(20)		MCWB	54.5	52.0	56.3	60.5	65.5	70.5	69.1	72.1	66.7	64.4	59.7	55.6	(20)	
(21)		2%	DB	55.4	55.7	62.6	69.5	75.5	82.8	85.9	84.2	79.4	69.1	61.7	56.9	(21)
(22)		MCWB	53.5	51.6	53.7	58.4	63.4	67.1	67.2	67.6	64.7	62.4	58.5	54.7	(22)	
(23)		5%	DB	54.0	53.9	58.6	65.1	71.0	78.7	81.6	79.2	75.4	66.4	60.0	55.7	(23)
(24)		MCWB	52.1	51.0	52.1	55.8	61.1	65.1	66.0	66.1	63.2	61.0	57.2	53.6	(24)	
(25)		10%	DB	52.8	52.5	55.8	61.3	67.3	74.7	77.0	75.2	71.8	64.5	58.3	54.6	(25)
(26)		MCWB	50.8	50.0	51.1	54.1	59.4	63.5	64.7	64.5	62.1	60.3	55.5	52.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	54.0	57.3	62.8	67.8	72.3	71.6	73.7	69.3	65.7	61.2	56.6	(27)
(28)		MCDWB	56.2	56.4	65.3	71.1	78.1	84.8	84.6	87.1	79.1	70.7	62.5	57.5	(28)	
(29)		2%	WB	54.0	52.8	55.1	59.5	64.5	68.9	69.2	70.1	67.0	63.8	59.4	55.2	(29)
(30)		MCDWB	55.0	54.6	59.9	67.4	73.0	79.1	80.6	80.4	75.1	67.2	61.0	56.4	(30)	
(31)		5%	WB	52.6	51.8	53.7	56.9	62.1	66.7	67.6	67.8	65.4	62.5	57.7	54.1	(31)
(32)		MCDWB	53.8	53.4	57.0	63.3	69.1	76.1	77.7	75.7	71.9	65.3	59.8	55.5	(32)	
(33)		10%	WB	51.1	50.5	52.3	54.9	60.2	64.6	66.1	66.0	63.8	61.1	55.8	52.8	(33)
(34)		MCDWB	52.5	52.3	55.2	60.2	66.2	72.7	74.3	72.2	69.5	63.8	58.1	54.4	(34)	
(35)	Mean Daily Temperature Range		MDBR	10.4	12.2	14.0	15.8	16.1	17.5	17.3	17.7	18.5	14.4	12.3	11.0	(35)
(36)		5% DB	MCDBR	9.4	12.7	19.2	22.2	22.6	24.5	26.2	25.3	24.0	16.1	11.9	9.4	(36)
(37)			MCWBR	8.2	9.9	12.4	12.9	12.3	11.4	10.9	11.4	11.6	10.5	9.6	8.6	(37)
(38)			MCDBR	8.8	10.9	14.5	19.5	19.8	21.8	21.2	20.1	19.0	13.5	10.8	9.1	(38)
(39)		5% WB	MCWBR	8.4	9.4	10.0	12.1	11.5	11.3	10.3	10.4	10.8	9.8	9.7	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.348	0.376	0.392	0.398	0.401	0.395	0.384	0.367	0.365	0.341	0.327	(40)	
(41)		taud	2.423	2.393	2.317	2.280	2.313	2.337	2.355	2.404	2.430	2.432	2.433	2.403	(41)	
(42)		Ebn at Noon	235	257	266	273	275	274	274	273	267	247	230	222	(42)	
(43)		Edh at Noon	22	28	35	39	39	39	37	34	30	26	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	344	639	1020	1511	1787	1967	1882	1656	1314	754	434	301	(44)	
(45)		RadStd	32	65	107	151	124	173	89	119	90	60	38	34	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page